

# SOLUTION REQUIREMENTS

|               |                                                       |
|---------------|-------------------------------------------------------|
| DATE          | 29 September 2026                                     |
| TEAM ID       | SWTID-2026-1812                                       |
| PROJECT NAME  | Comic craft Ai comic story creator using gemini model |
| MAXIMUM MARKS | 4 Marks                                               |

The Solution Requirements define the functional and technical requirements of the proposed ComicCraft – AI Comic Story Creator system. The requirements describe the features needed to generate a complete AI-assisted comic from a user-provided story prompt.

## 1. Functional Requirements

| Requirement Category | Requirement Description                                                                                   | Priority |
|----------------------|-----------------------------------------------------------------------------------------------------------|----------|
| User Input           | The system shall allow users to enter a story prompt for creating a comic.                                | High     |
| Character Input      | The system shall allow users to provide the main character name and basic character details.              | High     |
| Setting Selection    | The system shall allow users to select or specify a story setting such as forest, school, space, or city. | Medium   |
| Tone Selection       | The system shall provide options such as funny, dramatic, poetic, and light-hearted for the story tone.   | Medium   |
| Art Style Selection  | The system shall allow users to select an appropriate                                                     | Medium   |

|                          |                                                                                                         |        |
|--------------------------|---------------------------------------------------------------------------------------------------------|--------|
|                          | comic art style such as anime, pixel art, comic book, or realistic.                                     |        |
| Story Outline Generation | The system shall generate a structured five-panel comic story outline using Gemini AI.                  | High   |
| Story Narration          | The system shall generate narration and dialogue for each comic panel using AI.                         | High   |
| Image Generation         | The system shall generate suitable images for the comic panels based on the generated story content.    | High   |
| Comic Preview            | The system shall display the generated panels, narration, dialogue, and images in a comic preview page. | High   |
| Panel Management         | The system shall organize the generated content into separate comic panels in the correct sequence.     | High   |
| PDF Export               | The system shall allow the generated comic to be exported as a downloadable PDF file.                   | High   |
| Error Handling           | The system shall display an appropriate message when an input or AI generation process fails.           | Medium |

## 2. AI and Processing Requirements

| Requirement Category | Requirement Description                                                                          | Priority |
|----------------------|--------------------------------------------------------------------------------------------------|----------|
| AI Story Processing  | The system shall process the user's story prompt and convert it into a structured comic concept. | High     |

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|-------------------------|-----------------------------------------------------------------------------------------------------------|------|
| AI Outline Generation   | Gemini AI shall be used to generate the basic five-panel story structure.                                 | High |
| AI Narration Generation | Gemini AI shall generate suitable narration and dialogue for the comic panels.                            | High |
| Image Processing        | The system shall convert panel descriptions into suitable comic images using an image-generation service. | High |
| Content Integration     | The system shall combine the generated story, dialogue, narration, and images into a single comic output. | High |

### 3. Non-Functional Requirements

| Requirement Category | Requirement Description                                                                                 | Priority |
|----------------------|---------------------------------------------------------------------------------------------------------|----------|
| Usability            | The application shall provide a simple and user-friendly interface for creating comics.                 | High     |
| Performance          | The system should process user requests and display results within a reasonable time.                   | Medium   |
| Reliability          | The system should handle generation failures without crashing the application.                          | High     |
| Availability         | The application should be accessible whenever the required application services and APIs are available. | Medium   |
| Maintainability      | The application code shall be organized into separate                                                   | Medium   |

|                |                                                                                                             |        |
|----------------|-------------------------------------------------------------------------------------------------------------|--------|
|                | modules for easier maintenance and updates.                                                                 |        |
| Scalability    | The system should allow additional AI models and features to be integrated in the future.                   | Low    |
| Compatibility  | The web application should work with commonly used modern web browsers.                                     | Medium |
| Security       | API keys and sensitive configuration values shall be stored securely and not exposed in the user interface. | High   |
| Data Handling  | User-provided story information shall be processed appropriately for generating the requested comic.        | Medium |
| Output Quality | The generated comic should maintain a clear connection between the story, narration, dialogue, and images.  | High   |